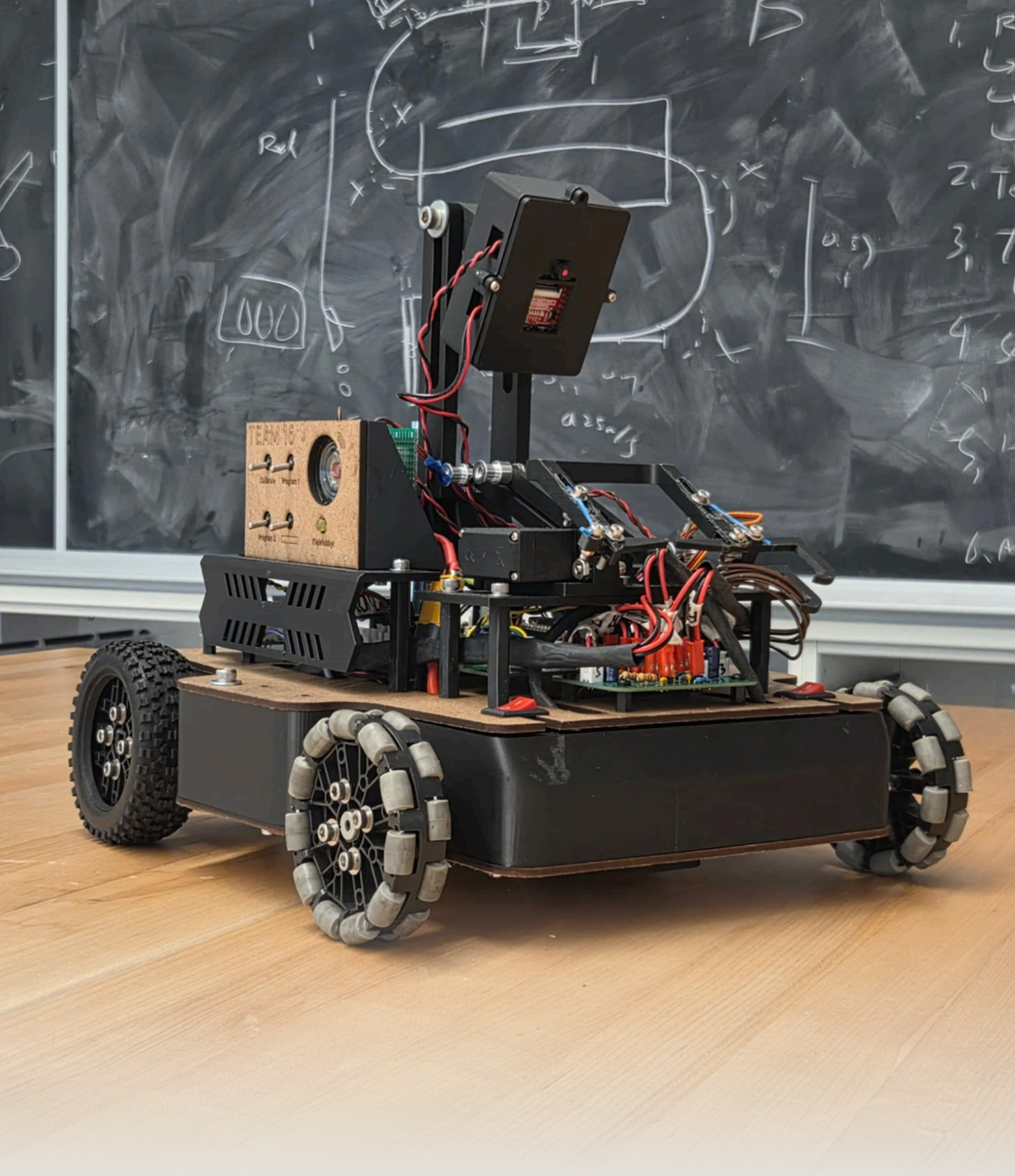
hi I'm Morgan

Come look at my projects!

2026

UBC ENGINEERING PHYSICS

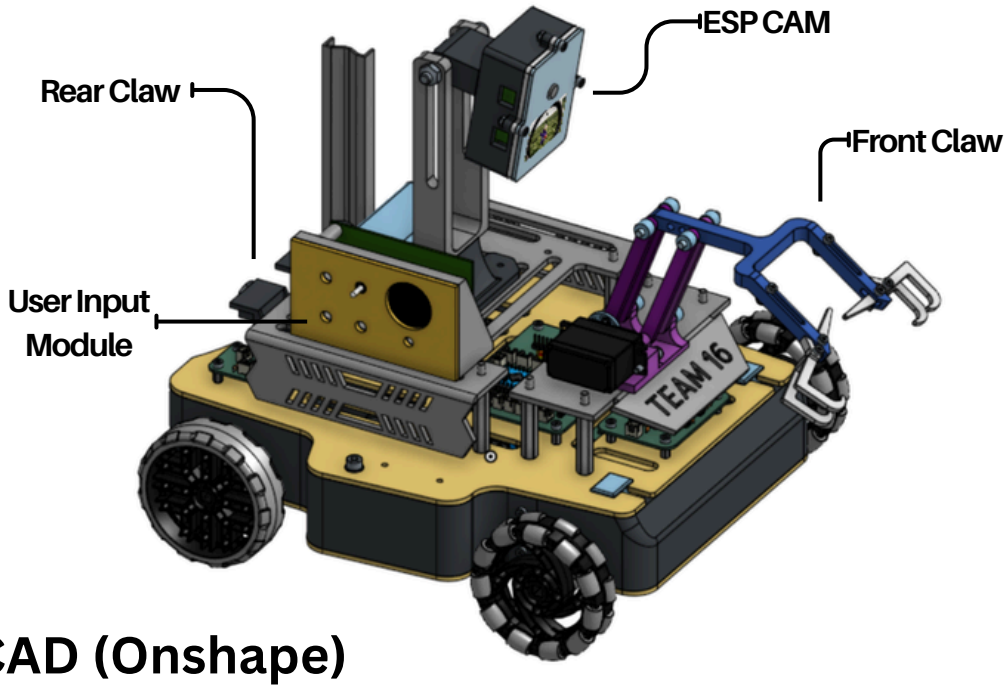
morganchangmc@gmail.com | Resume | LinkedIn



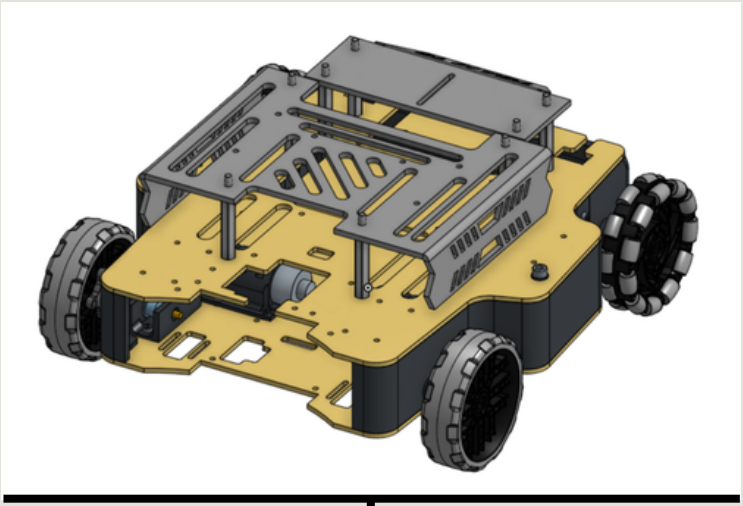
Autonomous Robot

ENPH 253 | MAY - AUG 2026

Robotics, PCB Design, Mechanical Design



Detailed CAD (Onshape)

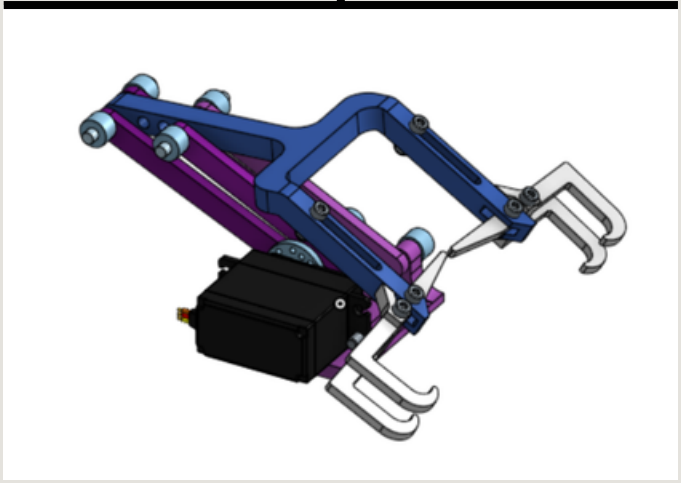


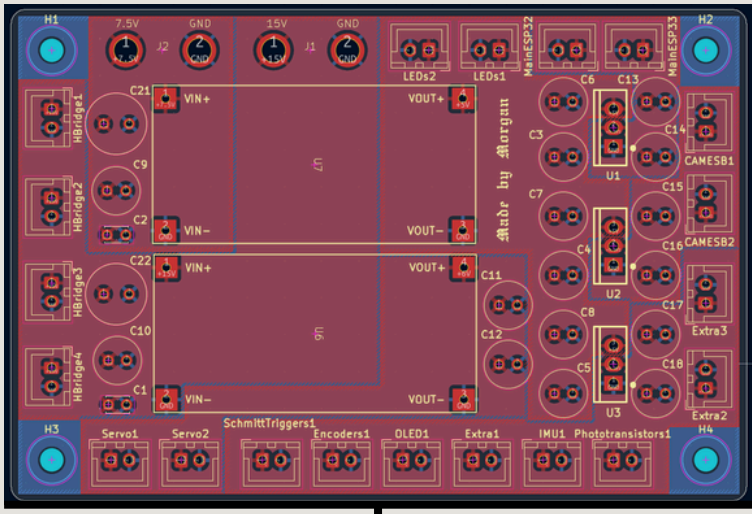
Drivetrain

Designed a differential drive with an adjustable gearing belt drive system

Designed and tested two end effectors for picking up different objects

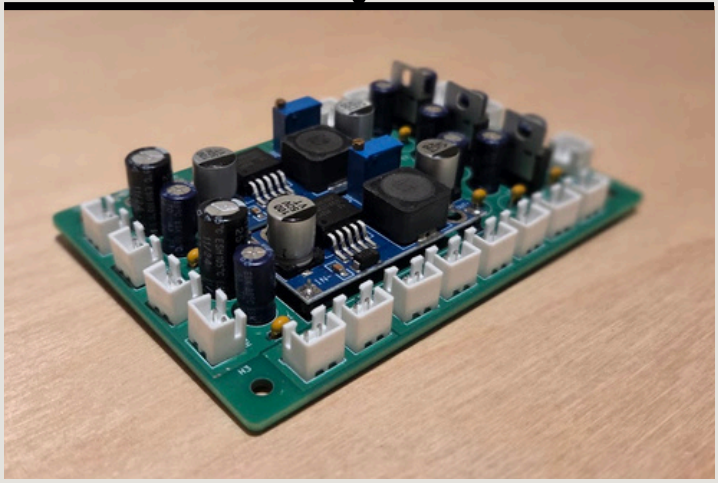
End Effectors





PCB Design + Testing

Soldered and tested a system of 8 PCBs for motor control, sensor input, microcontrollers



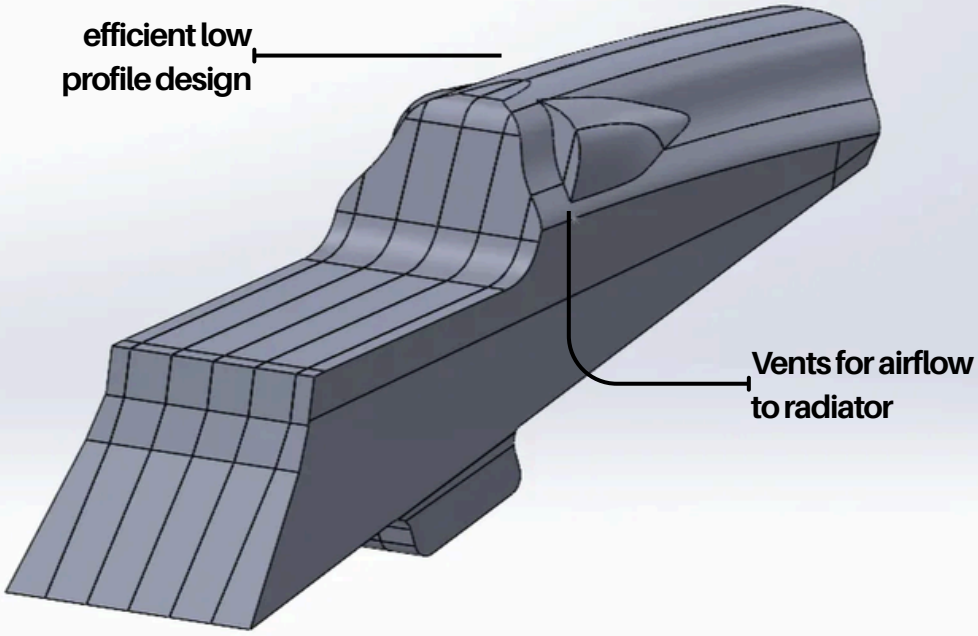
Power Distribution Board



Motorcycle Tail Fairing

UBC Thunderbikes | SEPT 2024 - MAY 2025

Composites, CNC Machining



Mould CNC Milled from Foam



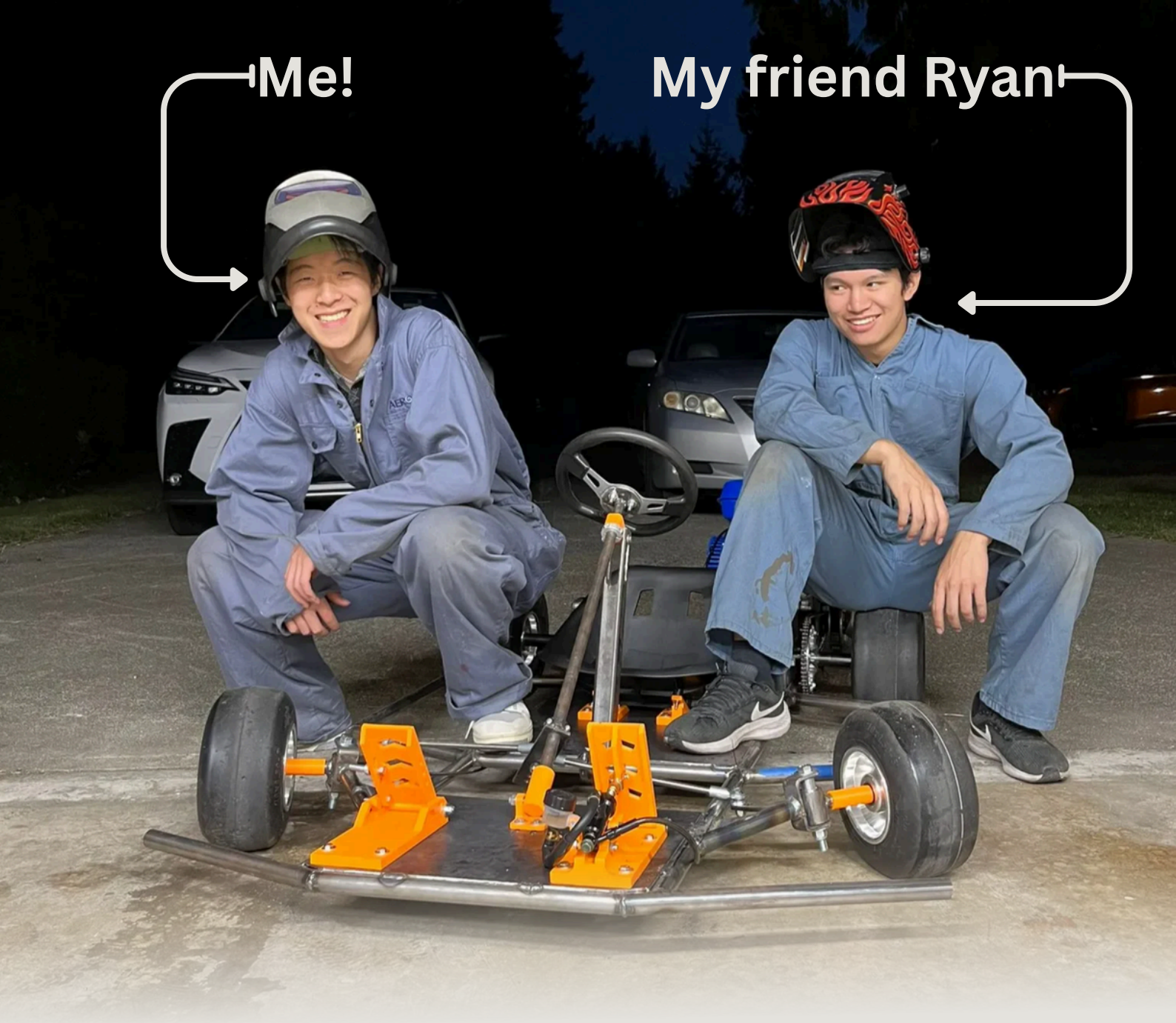
The mould was milled in 6 pieces (shown in CAD) from a foam block on a CNC router

Rightmost (outside) piece shown



We used a wet layup method, cured under vacuum

Untrimmed , post cure part shown



Go-Kart

Personal | JUL - AUG 2025

Mechanical Design, MIG Welding



Running Go-Kart

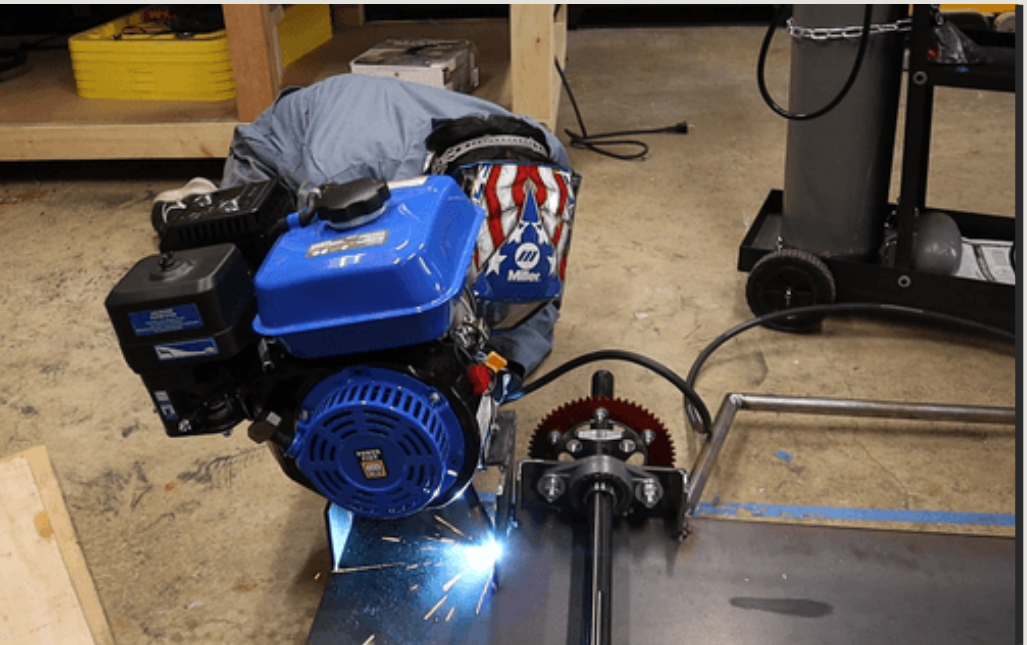
Engine: 300cc Powerfist single (~7 hp)
Top speed: ~50km/h
Transmission: CVT
Brakes: Single caliper locked to rear axle

Custom 3D Printed Parts

Brake pedal, throttle pedal, seat mounts, and steering mounts were designed, printed and assembled



Brake pedal shown (very safe)



MIG Welded

Trellis chassis, engine mount, and floor plates safely (kinda) welded by two guys who have never welded before

Me welding the engine mount